

Unsupervised Deep Learning for Ecological Site Groups in MLRA 42B

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Problem Statement

Across the western United States, land managers, ranchers, and conservation agencies rely on a system of land classification called Ecological Sites. An Ecological Site is a patch of landscape with a distinct combination of soil, climate, and topography that determines what plants can grow there and how the land responds to disturbance. It describes the whole soil-climate-vegetation system and how it can shift between different states over time (Karl and Herrick, 2010; Bestelmeyer et al., 2003). Ecological site maps are built largely by hand: soil scientists and ecologists visit sites, take measurements, and use expert judgment to draw boundaries and assign classifications. This is accurate but slow and expensive, and in practice ecological site maps are produced by linking ecological sites to soil survey (SSURGO) map units rather than being mapped independently.

The ecological site concept has become cumbersome, many sites with different names have properties that are functionally quite similar. The USDA would like to transition from using ecological sites to the ecological site group, effectively absorbing ecological sites with similar properties into broader classification categories. This project asks whether an unsupervised deep learning model can divide MLRA 42B (Southern Rio Grande Rift) into ecologically meaningful groups using only environmental covariates. We then ask where those groups agree and disagree with mapped ecological sites. Disagreement is the main interest, it may give insight into existing map-unit uncertainty, ecological site concepts that don't fit the landscape, or gaps in our covariates or model.

Motivation & Objectives

Deep learning fits this problem for three reasons. The inputs are high-dimensional and multi-source. Ecologically important patterns are spatial (landform position, drainage, soil properties) and are lost when pixels are treated independently (Padarian et al., 2019; Wadoux et al., 2019). The goal is to discover groups without labels, which suits representation learning. Prior digital ecological-site mapping has been supervised and trained on existing ESG classes (Maynard et al., 2019). Our unsupervised approach does not assume current ecological site concepts capture all the heterogeneity of landscape patterns.

The task type is unsupervised clustering

Objectives:

1. Use a convolutional autoencoder to condense the stacked landscape data and capture its key environmental and spatial patterns.
2. Define the number of clusters and produce a map of ecological groups.
3. Describe each group by the range of its covariates.
4. Measure agreement with SSURGO-mapped ecological sites and characterize where and why they disagree.

5. Depending on time, test whether groups differ in vegetation response to drought using Rangeland Analysis Platform (RAP) cover time series, held out as independent validation.

Dataset Description

All data are public. Model inputs and validation data are kept separate. The study extent is the MLRA 42B boundary.

Layer	Role	Source	Native resolution / format	Rationale for inclusion
Climate normals	Input	PRISM 30-yr normals	~800 m raster	Precipitation, the timing of precipitation, solar exposure, and average temperature have a large impact on what vegetative community can survive in each location
Elevation (slope, northness/eastness, curvature)	Input	USGS 3DEP DEM	10 m raster	Particularly in the arid southwest, elevation has a large influence on the temperature and precipitation. The aspect has an influence on solar radiation and can influence plant communities Slope and other elevation derived metrics have large influence on the soil properties of each site
Spectral (seasonal medians, NDVI)	Input	Landsat 8/9 surface reflectance via Google Earth Engine	30 m raster	The reflectance of different plant types gives insight into varying plant communities. Bare ground reflectance gives insight into the mineralogy of the soil.
Geology (lithology classes)	Input	USGS State Geologic Map Compilation	Vector polygons	This has an influence on the soil, specifically the soil parent material. This will add some context to the mineralogy of the soil in a given location.
Ecological sites	Validation	SSURGO components linked to ESDs (EDIT)	Vector map units + tables	This will be used after the model has run to compare the output map to an existing map

Layer	Role	Source	Native resolution / format	Rationale for inclusion
				ecological site. The variance between the two is the interesting component.
Vegetation cover time series	Validation	Rangeland Analysis Platform (RAP)	30 m annual raster	This looks at rangeland specific responses to a type of disturbance; this map should also be comparable to the output and will be interesting to compare for validation.

Preprocessing thoughts:

- Put all layers on the same 30 m grid. For continuous data like elevation or climate, we'll estimate new values by averaging nearby cells. For category data like rock type, we'll copy the value of the closest cell so categories don't get blended. Climate data comes at a much coarser resolution (800 m).
- The elevation data and reflectance will need to be stitched together, masked using the MLRA 42b bounding area.
- Turn the geology map into a format the model can read. We'll convert the geology polygons into a grid. Then each rock type becomes its own one-hot encode

Proposed Methodology

Baseline: PCA on the per-pixel covariate stack, then k-means.

Deep learning model: Convolutional neural network, specifically a convolutional autoencoder.

How I understand it to work:

1. The encoder (3–4 convolution + pooling blocks) compresses each patch into a latent vector of about 16–32 dimensions.
2. The decoder mirrors the encoder and reconstructs the patch. Training minimizes reconstruction loss, so no labels are needed.
3. We cluster the latent vectors with k-means and a Gaussian mixture model. We will also try joint clustering with Deep Convolutional Embedded Clustering (Guo et al., 2017), which refines the clusters during training.
4. Each pixel takes the cluster of the patch centered on it, which produces the prediction map.

Choosing the number of clusters: Need to do a little more research on how you make these determinations. Ideally, we choose the number of clusters to achieve the most ecologically meaningful clusters, I am not sure how to assess this now.

Evaluation:

- Agreement with ecological sites: We will compare two rules: dominant component only, and weighting by component percentage.
- Disagreement: map disagreement hotspots and relate them to map-unit complexity, landform, and covariate uncertainty.
- (Time dependent) Response validation: compare RAP perennial and annual cover response to a documented drought year across groups. Groups that are truly functional should differ more between each other than within.

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